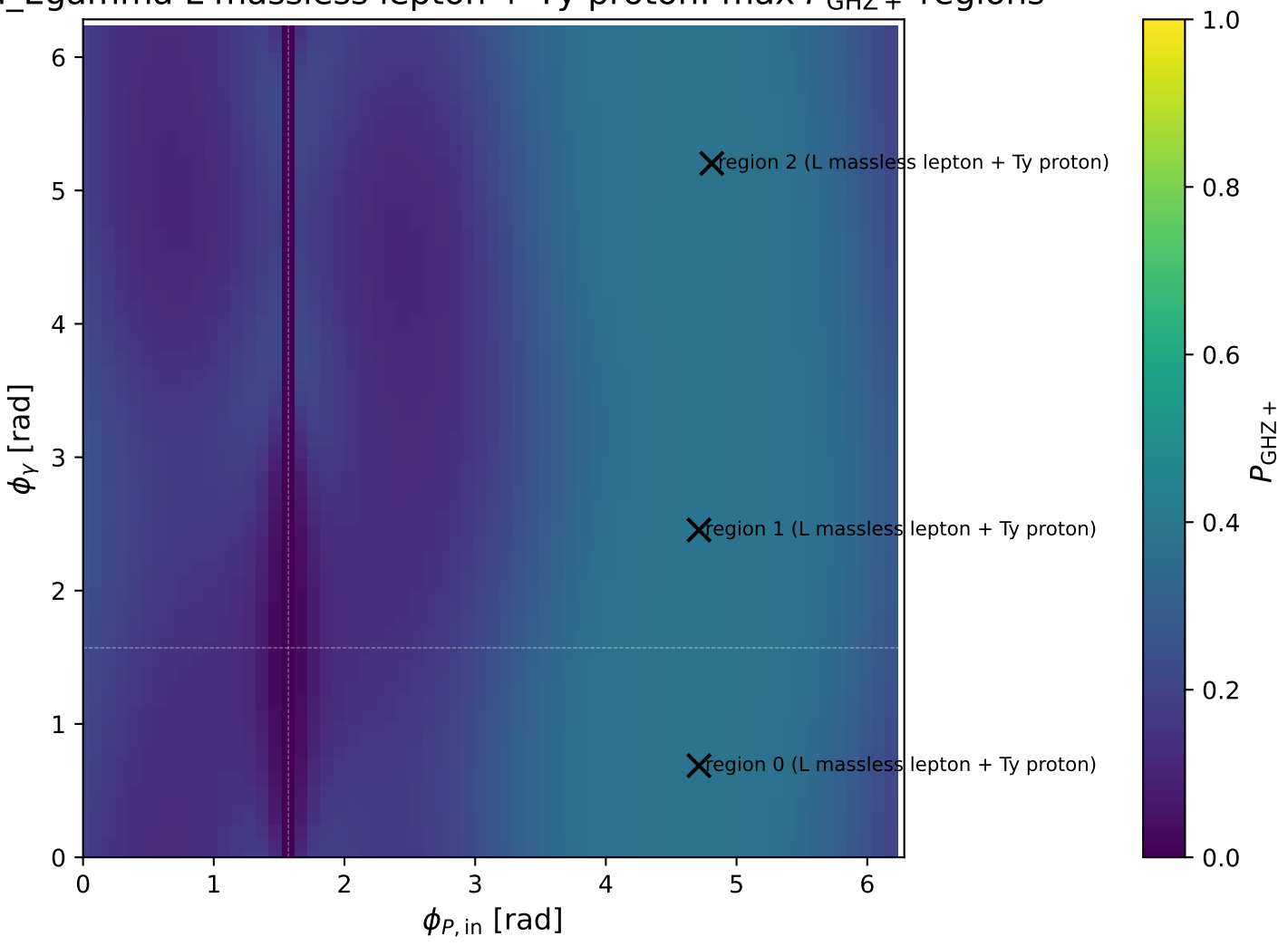
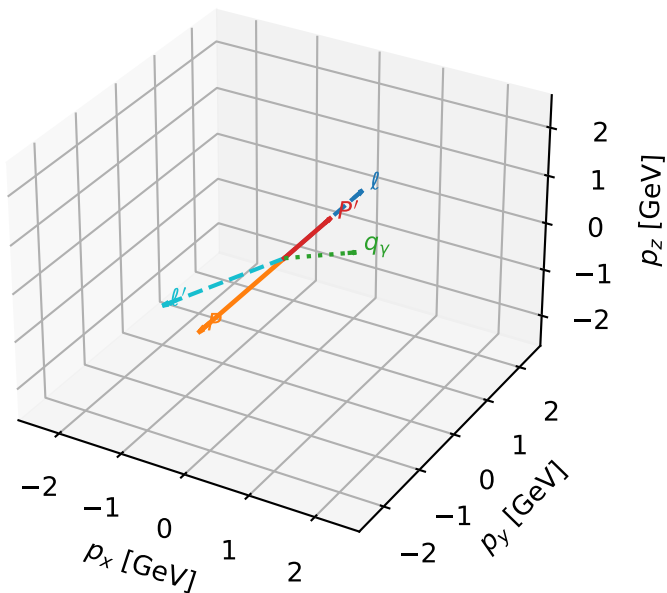


I_Egamma L massless lepton + Ty proton: max P_{GHZ} + regions



L massless lepton + Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

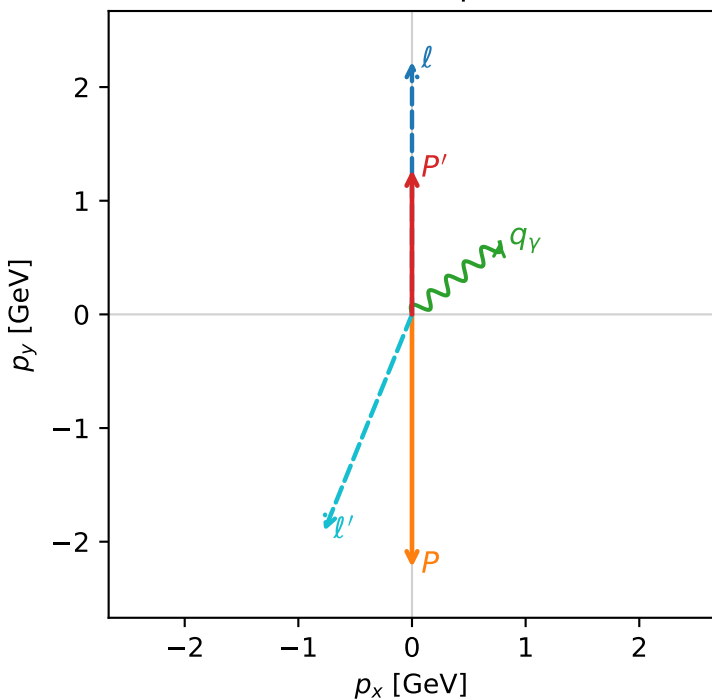


ghz_purity_mid_egamma_lty_region_0 $P_{\{\text{rm GHZ+}\}}=0.384124$
 region: mid_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

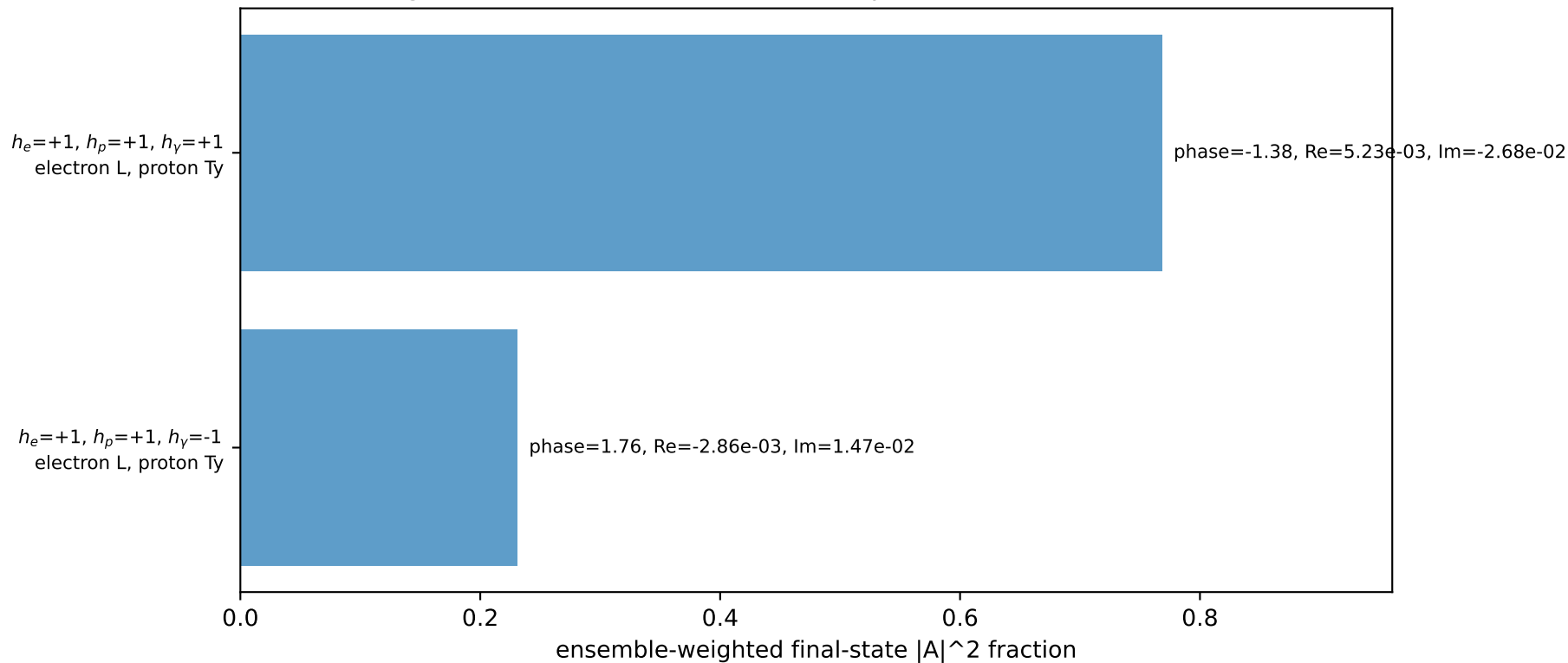
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.27253$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_p=4.71239$
 $E_{\gamma}=1$, $\phi_{\gamma}=0.687223$
 $Q^2=17.6377$, $x_B=0.919354$, $t=-11.5344$, $W^2=2.42703$, $y=0.92968$
 $\theta(\ell', \gamma)=151.441$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.0576$ $p_x= -0.7730$ $p_y= -1.9069$ $p_z= 0.0000$
 P' $E= 1.5809$ $p_x= 0.0000$ $p_y= 1.2725$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

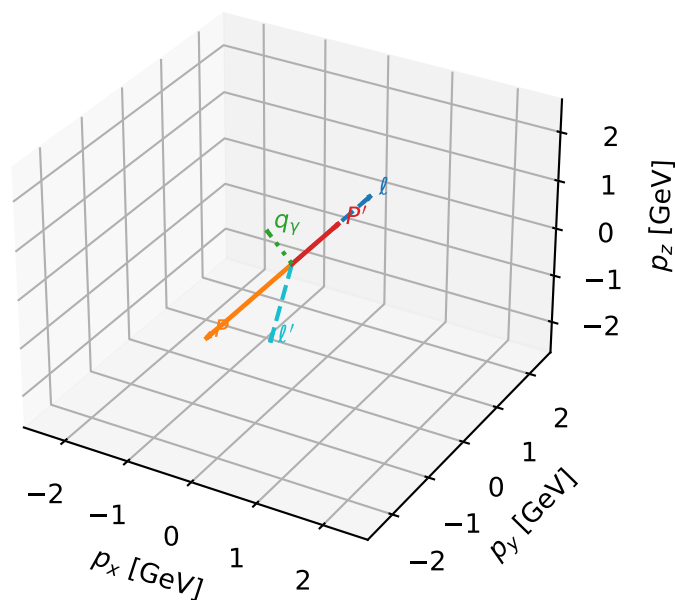


Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton + Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

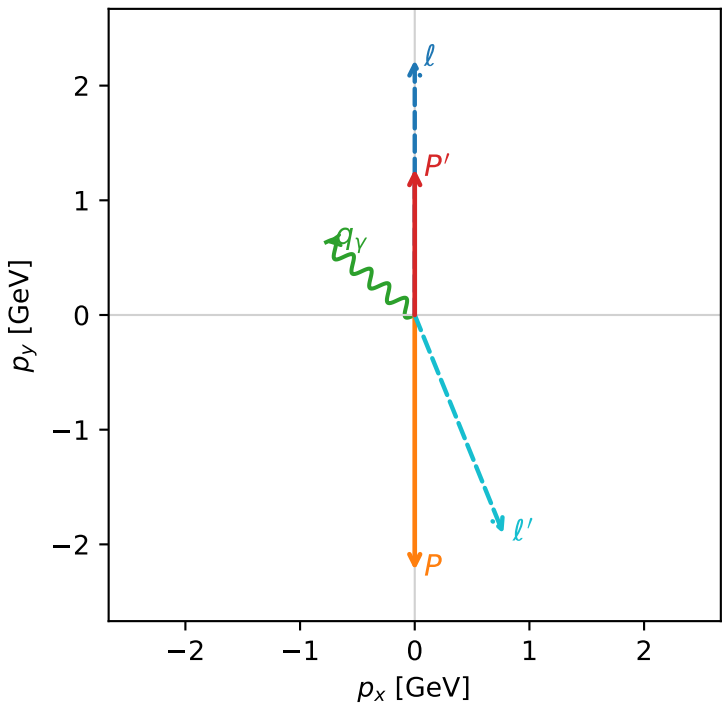


ghz_purity_mid_egamma_lty_region_1 $P_{\{\text{rm GHZ+}\}}=0.384124$
region: mid_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

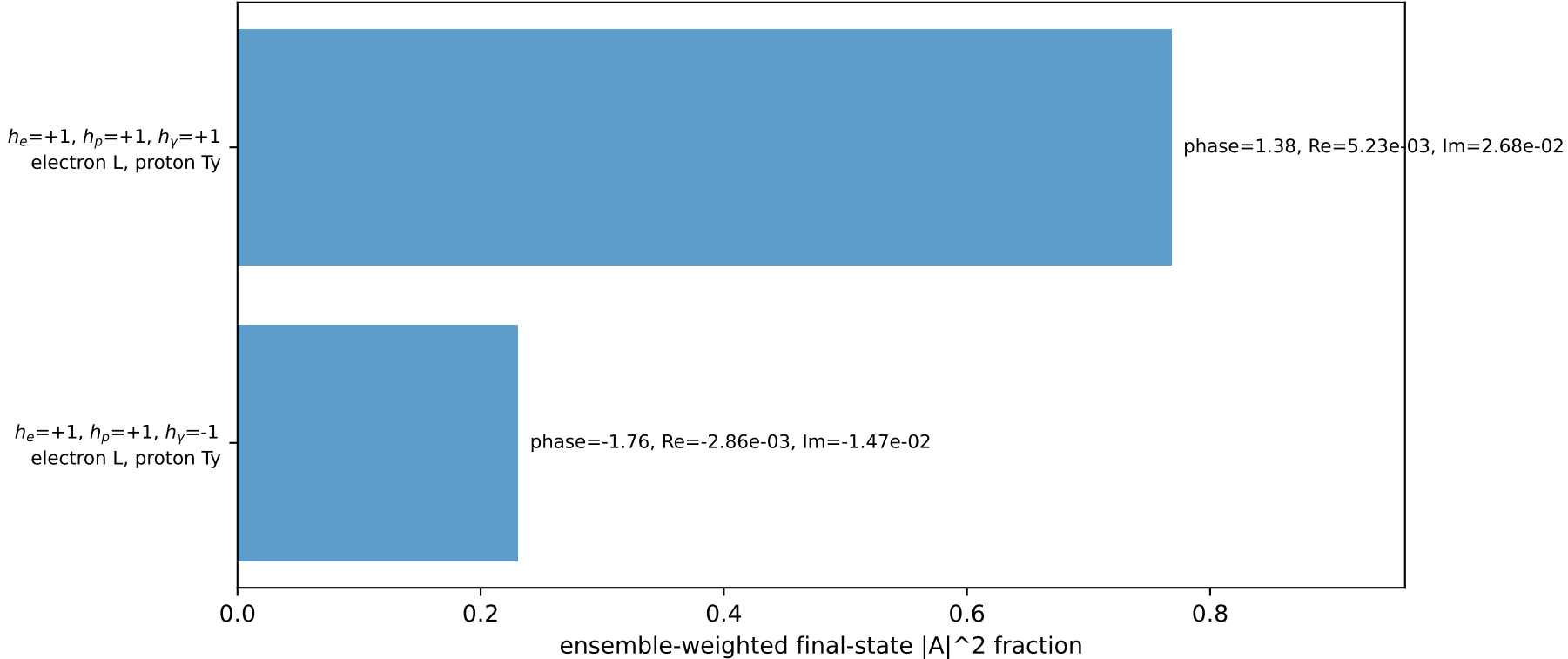
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.27253$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_p=4.71239$
 $E_Y=1$, $\phi_Y=2.45437$
 $Q^2=17.6377$, $x_B=0.919354$, $t=-11.5344$, $W^2=2.42703$, $y=0.92968$
 $\theta(\ell', \gamma)=151.441$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.0576$ $p_x= 0.7730$ $p_y= -1.9069$ $p_z= 0.0000$
 P' $E= 1.5809$ $p_x= 0.0000$ $p_y= 1.2725$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

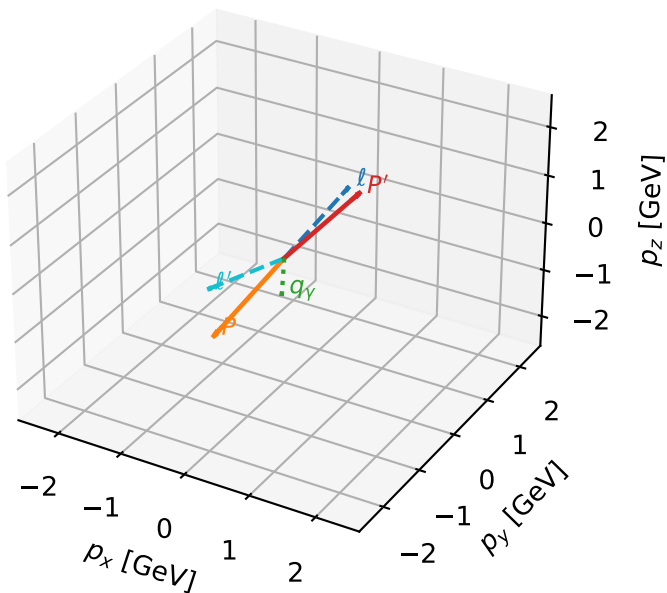


Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton + Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta



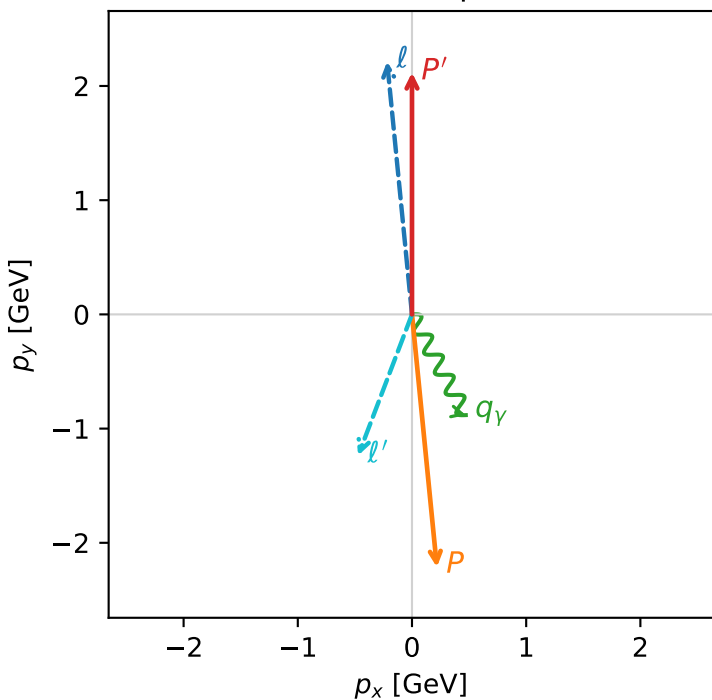
ghz_purity_mid_egamma_lty_region_2 $P_{\{\text{rm GHZ+}\}}=0.383919$
 region: mid_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{l_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.11756$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$
 $E_\gamma=1$, $\phi_\gamma=5.20326$
 $Q^2=11.1487$, $x_B=0.571267$, $t=-18.7978$, $W^2=9.24691$, $y=0.945717$
 $\theta(l', \gamma)=49.0069$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2244$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 l' $E= 1.3225$ $p_x= -0.4714$ $p_y= -1.2356$ $p_z= 0.0000$
 P' $E= 2.3160$ $p_x= 0.0000$ $p_y= 2.1176$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.4714$ $p_y= -0.8819$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.9%)

